

Unveiling the Use of Ontologies in Adaptive User Interface Systems Development: A Systematic Mapping Study

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Abstract. Ontologies have been used in software development to aid in several contexts. Aiming to investigate whether and how ontologies have been used to develop AUI systems, we performed a systematic mapping study (SMS). 165 publications were identified, of which 30 address AUI system development with the aid of ontologies. After grouping complementary works, 23 proposals were analyzed. Twelve research questions were defined to investigate the distribution of publications over the years and types of vehicles, the research types, and the following aspects: (i) the use of ontologies to aid in developing AUI systems and (ii) the used ontologies. As a result, we noticed that ontologies have primarily been used to provide a conceptual foundation for organizing elements like user profiles, devices, and context of use, supporting dynamic UI adaptations mainly in the implementation phase. Most ontologies are operational, with OWL as the predominant language, covering domains like Context of Use, Devices, and User Profile. Reusing other ontologies has not been prominent, but some works have done so, and the use of fundamental ontologies to ground the developed ontologies has not been a concern. The results highlight the need to explore systematic ontology-based methods that support the reuse and application of ontologies in the development of AUI systems.

Keywords: Adaptive User Interface, Ontology, AUI System, Systematic Mapping Study

1. Introduction

The digital age has brought a growing demand for interactive systems that are not only intuitive but also better aligned with user needs. A critical component to achieve this is a well-designed User Interface (UI), which plays a pivotal role in enabling effective communication between users and systems. The quality of the UI directly impacts the success of any interactive system. However, the increasing diversity of electronic devices, user environments, and individual needs introduces significant challenges in designing and developing interactive systems (Sebek et al., 2015).

Adaptive User Interfaces (AUIs) have emerged as a solution to these challenges. An AUI is an interface designed to adjust itself (e.g., layout, content, and functionality) based on various factors (e.g., user

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preferences, device capabilities, context of use, and environmental conditions) (Soui et al., 2017). These interfaces aim to provide personalized and distinctive user experiences tailored to individual users, addressing the broad spectrum of needs and environments in which they operate (Iqbal et al., 2018).

The rapid technological advancements of recent decades have also given rise to products and services that require specialized user skills. In the context of ubiquitous computing, individuals with disabilities often face unique challenges, highlighting the critical importance of prioritizing accessibility in system interfaces (Zouhaier et al., 2015). This underscores the need for generic yet adaptable solutions that address the requirements of all users, particularly those with disabilities.

Developing AUI systems necessitates managing a substantial body of knowledge. This includes understanding both the characteristics of the system's UI and the specific needs of its users. Ontologies have proven to be effective tools for addressing such knowledge-related challenges (Feilmayr and Wöß, 2016). Defined as a formal and explicit specification of a shared conceptualization (Gruber, 1995; Studer et al., 1998), ontologies facilitate knowledge structuring, semantic data interpretation, and reasoning. They are widely used in conceptual description, communication, knowledge management, semantic web applications, and interoperability solutions (Taye, 2010).

Ontologies can function as conceptual models (reference ontologies) or computational artifacts (operational ontologies) that are interpretable by machines, enabling automated solutions (Guizzardi, 2007). The field of AUI has particularly benefited from ontologies in various contexts, including automatic adaptation and real-time interface adjustments. For instance, Bonacin et al. (2022) explore how ontologies enhance adaptability and usability in AUI development, while Braham et al. (2022) examine how ontologies model user preferences and behaviors to create more intuitive interfaces. Other studies, such as Fedasyuk and Lutsyk (2021), Stefanidi et al. (2022), and Sala et al. (2021) address the use of ontologies in real-time systems and automated adaptation, emphasizing their role in personalization and enhanced user experiences.

Aiming to further investigate how ontologies have been used to aid AUI system development, we conducted a systematic mapping study (SMS). SMS provides a structured and auditable method for exploring and classifying existing research within a specific field (Kitchenham and Charters, 2007). We aimed to provide a panorama about the topic, mainly regarding how ontologies have been used to support AUI development and characteristics of the ontologies employed.

In summary, the results showed that ontologies are primarily used to define and organize AUI system components. They are also used to support dynamic UI adaptations through semantic reasoning. Most of the analyzed ontologies prioritized defining UI adaptation rules directly in the system instead of embedding them in the ontology. Concerning the used ontologies, we identified 23 ontologies addressing various AUI-related domains, including UI, user profiles, devices, and usage contexts. These ontologies have been utilized both as conceptual models and computational artifacts, serving purposes such as knowledge structuring, semantic reasoning, and supporting AUI development. While operational aspects were a key focus, the reuse of existing ontologies and the use of foundational ontologies have not been a concern. Frequently addressed domains include Context of Use, Devices, and User Profiles, whereas Adaptive UI and Design Patterns are underrepresented. Despite their contributions, many ontologies exhibit significant limitations, such as incomplete documentation and implicit details, which hinder their understanding and broader application.

Before conducting this study, we looked for secondary studies investigating the topic, and we did not find any. Given the absence of previous secondary studies on this topic, our research is pioneering in providing a comprehensive investigation into the role of ontologies in AUI development.

The study provides a panorama on the use of ontologies in AUI system development and offers valuable insights for researchers and practitioners in the fields of AUI systems and ontologies (including system engineers, developers, designers, ontology engineers, among others). It provides an overview of existing ontologies and their applications, contributing to reuse and inspiring new ontology-based solutions. For example, designers can reuse ontologies addressed in the selected studies for user modeling, context representation, and UI adaptation, and use existing solutions as examples to develop AUI systems. Ontology engineers, in turn, can reuse, integrate, and evolve existing ontologies to support AUI system developers. Researchers can explore the identified gaps in future studies, advancing the state of the art in AUI systems and ontologies. Moreover, the study contributes to bringing the ontology and AUI system communities closer, showing some benefits of this integration and challenges that still need to be overcome.

This paper is organized as follows: Section 2 provides the background for the study and discusses related work. Section 3 presents the research protocol. Section 4 outlines data extraction and synthesis. Section 5 discusses the findings. Section 6 addresses the study limitations. Finally, Section 7 concludes the paper.

2. Background

2.1. Adaptive User Interface (AUI)

In contemporary interactive systems, the demand for adaptive designs that prioritize human needs and preferences has significantly increased. A well-known and currently discussed challenge in this field pertains to usability problems that arise when diverse users¹ groups interact with the same system (Rathnayake et al., 2019).

Users exhibit significant variation across demographic factors, educational backgrounds, cognitive abilities, and personal preferences (Alves et al., 2020). Beyond these characteristics, users' motivations, moods, and goals also fluctuate dynamically. This variability requires interactive system interfaces that adapt not only to user preferences but also to contextual factors, ensuring that systems are inclusive of groups such as the elderly and individuals with disabilities (Castillejo et al., 2014a). Ubiquitous computing scenarios, for instance, frequently overlook the needs of individuals with disabilities, underscoring the importance of designing interfaces that promote inclusivity (Zouhaier et al., 2015).

Interactive systems inherently incorporate a user interface (UI), serving as the primary medium for user interaction with technology (Farooqui et al., 2021). The International Organization for Standardization (ISO) ISO (2019) defines a user interface as all the interactive system's components that allow interactions between humans and machines to occur, and emphasizes that the design of such interfaces should facilitate effective interaction by providing mechanisms for users to input data and receive feedback. Hussain et al. (2018) define UI as the interface connecting the user to a system or device.

UI personalization can be user-driven or system-driven and is represented by adaptable and adaptive UIs (Eichler and Dostál, 2012). Adaptable UI refers to the ability of users to explicitly customize the UI (i.e., it enables end users to modify the interface by adjusting table columns, managing data entry fields, or altering the UI's appearance) to meet the specific requirements of your tasks (Rivera, 2005). In contrast, an adaptive UI is the practice of adjusting the layout or elements of a UI based on user

¹ According to ISO/IEC ISO (2024) standard, diverse users means individuals with differing abilities and characteristics or accessibility needs.

needs (Schneider-Hufschmidt et al., 1993). Adaptive UIs automatically modify UI objects and layout to suit a group of users, enabling them to complete tasks (Zhang et al., 2020). In summary, adaptive UI dynamically and automatically adjusts the interface to conform to the user's needs and usage patterns, while adaptable UI relies on explicit user input to make adjustments. (Eichler and Dostál, 2012).

Enhancing UX and usability becomes a concern in the development of such systems. These systems can also exhibit adaptability, i.e., the capability to automatically adjust to changes in their environment or context of use (Yigitbas et al., 2020). It is feasible to direct this adaptation specifically toward the UI, resulting in an AUI. The goal of AUI is to enhance User Experience (UX) and usability by dynamically modifying the system's behavior according to the context or specific user characteristics (Yigitbas et al., 2020).

AUI is distinct from a static UI, which remains fixed and does not change based on user interaction or environmental shifts (Oppermann, 1994; Yigitbas et al., 2020; Langley, 1999). This real-time adaptability enhances usability and accessibility by ensuring that interfaces align with individual user needs during execution (Langley, 1999). Therefore, UI is considered adaptive when it automatically adjusts itself based on the user's characteristics or environment.

A key factor in the development of AUIs is the challenge of designing systems that cater to diverse user characteristics, such as cognitive abilities, learning styles, preferences, and even physical abilities (Yigitbas et al., 2020). While systems are typically engineered to fulfill the requirements of the majority of users, involving people with disabilities in the design process, although challenging, is important to enhance understanding of their complexities and enable the creation of more meaningful and useful systems for them (Rupprecht et al., 2014). Developing an adaptive interface becomes pivotal in mitigating these challenges by considering diverse user characteristics and tailoring the interface to accommodate their needs, thereby enhancing system accessibility and usability (Weld et al., 2003).

Alongside this, the increasing complexity and efficacy of interactive systems, accessible across multiple devices, including computers, smartphones, tablets, and wearables, underscore the imperative for adaptive interfaces (Oppermann, 1994). In addition, the rise of technologies like wearables brings new challenges to computer system design and development (Al-Megren, 2018). These devices, with limited screen space and restricted interaction modalities (e.g., gestures, voice), typically reduce usability (Brewster et al., 2003). Users often prioritize their visual attention on navigating their environment, making visually demanding interface designs difficult to use (Brewster et al., 2003). AUIs can be tailored based on factors such as the user's experience level, task goals, and context of use (Akiki et al., 2014). By personalizing the user interface in alignment with these factors, AUIs provide users with a simpler experience when using the UI, contributing to the overall usability of software applications (Miraz et al., 2021).

According to Gullà et al. (2015), the design of an AUI involves: determining who should adapt and the role of the UI in the adaptation process; defining the primary goals to consider during adaptation; establishing a proper set of rules to manage the adaptation; identifying the level of interaction to focus on and the adaptation variables; and selecting methods for the adaptation process, including an inference mechanism for user choices. Designing an AUI is complex, as it requires addressing challenges such as assessing the user's mental state, defining appropriate adaptive behaviors for the UI, and ensuring adaptations that effectively respond to changes in user behavior or context (Yen and Acay, 2009).

Creating an AUI requires modeling information related to the user and the context of the interaction, which can be complex and require additional resources Akiki et al. (2014). However, research on AUIs continues to advance to provide highly usable systems for people with different needs and characteristics in different usage contexts Gullà et al. (2015).

2.2. Ontologies

According to Studer et al. (1998), an ontology constitutes a formal and explicit specification of a conceptualization. “*Conceptualization*” denotes an abstract and simplified portrayal of the world, identifying the relevant concepts of a phenomenon. The term “*explicit*” emphasizes the clear definition of the types of concepts used and the constraints imposed on their application. The descriptor “*formal*” signifies that the ontology must be machine-interpretable, excluding reliance on natural language. Lastly, the term “*shared*” embodies the notion that ontologies should encapsulate consensual knowledge accepted by a community.

Ontologies can be classified into three architectural layers (Scherp et al., 2011): (i) *Foundational Ontologies*: these underpin basic and universal concepts common to all domains, serving as a foundation for other ontologies and offering a general framework for knowledge representation; (ii) *Core Ontologies*: these concentrate on more specific and comprehensive concepts within a particular domain, potentially incorporating concepts from *Foundational Ontologies* to provide a detailed framework for knowledge representation; and (iii) *Domain Ontologies*: these narrow their focus to specific concepts within a defined domain, such as medicine or finance. They may utilize concepts from *Core* and *Foundational Ontologies*, furnishing an even more detailed framework for knowledge representation within the chosen domain.

In another classification, orthogonal to the one proposed by Scherp et al. (2011), Guizzardi (2007) differentiates between reference and operational ontologies. A *reference ontology* is a conceptual model whose aim is to make a clear and precise description of the elements (concepts, relations, properties) of a conceptualization for communication, learning, and problem-solving. A reference ontology is to be used in an offline manner to assist humans in tasks such as meaning negotiation and consensus establishment. Such a consensual conceptual model can be coded in a computational language (e.g., OWL) and then deployed in a computational environment. This implementation version is an *operational ontology*. Unlike reference ontologies, operational ontologies are not focused on representation adequacy but are designed to guarantee desirable computational properties.

In software development, both reference and operational ontologies play vital roles. Reference ontologies are typically employed during the software development phase to describe the application domain, known as *ontology-driven development* (ODD) (Happel and Seedorf, 2006). In contrast, operational ontologies emphasize computational properties and are frequently used during runtime to support application logic, forming the basis of *ontology-based architecture* (OBA) (Happel and Seedorf, 2006). In the Human-Computer Interaction (HCI) domain, ontologies have been used for several purposes. Some works (such as the ones by Storrs (1994), and Mezhouidi and Vanderdonckt (2015)) have used ontologies as a conceptual model to conceptualize the HCI domain (ODD). Others have used only operational ontologies (OBA). For example, in the work by Garay-Vitoria et al. (2019), an operational ontology about affective computing was used as a basis for developing Affective Interactive Systems. Some works take advantage of both reference and operational ontologies (ODD and OBA). Bakaev et al. (2019), for example, used a reference ontology about UI and UI measurement to structure and organize knowledge, and an operational version of the ontology was used in a tool to assess web UI using different metrics. These applications highlight how ontologies, whether reference or operational, are essential for structuring knowledge, guiding development processes, and enabling runtime functionality in HCI and software development.

3. Research Protocol

To perform our study, we followed the process defined by Kitchenham and Charters (2007), which comprises: planning, when the research protocol is defined to support study replicability as well as help researchers to avoid bias when conducting the study; conducting, when the protocol is executed and data is extracted, analyzed and recorded; and reporting, when the results are recorded and made available to potential interested parties. In this section, we present the protocol used in the study.

The study *goal* was to investigate how ontologies have been used to help develop AUI systems and the characteristics of the used ontologies. To achieve this objective, we formulated the *research questions* presented in Table 1.

Table 1: Research questions and their rationale.

RQ	Research Question	Rationale
RQ1	When and in which type of vehicle have the publications been published?	Provide an understanding of when and where (journal/conference/workshop) the publications have been published.
RQ2	Which types of research have been done?	Identify the nature of research according to the classification defined in Wieringa et al. (2006), which includes Evaluation Research, Proposal of Solution, Validation Research, Philosophical Paper, Opinion Paper, and Personal Experience Paper. This question is useful to evaluate the maturity stage of the research topic.
RQ3	What purposes have the ontologies been used for?	Identify the main purposes for utilizing ontologies in the development of AUI systems.
RQ4	In which software development phase have the ontologies been used?	Identify the phases in which ontologies have been employed (e.g., Requirements Specification, Modeling, Design, Implementation, Testing), and ascertain whether there has been a predominance of conceptual or implementing-related phases.
RQ5	How have ontologies been used to support the development of AUI systems?	Identify how ontologies have been utilized to aid in developing AUI systems and verify if similar or distinct approaches have been employed.
RQ6	What UI adaptations have been supported by the use of ontologies?	Identify the range of UI adaptations that have been based on ontologies.
RQ7	Have the AUI systems been developed following a systematic method? If so, has the method addressed the use of ontologies to develop the system?	Determine whether the development of AUI systems has followed well-defined steps and whether they guide the use of ontologies.

RQ	Research Question	Rationale
RQ8	Which types of ontology have been utilized to support the development of AUI systems?	Identify whether reference ontologies (conceptual models designed to provide a clear and precise description of entities in the domain of interest) or operational ontologies (those machine-readable and that deal with computational properties) have been used. This helps discern if the utilization of ontologies has predominantly centered on the conceptual or operational aspects of AUI system development.
RQ9	Which languages have been used to represent or implement the ontologies?	Determine the languages employed for representation (in the case of reference ontologies) or implementation (in the case of operational ontologies) and verify whether there is a prevailing technology or language.
RQ10	What domains have been covered by the ontologies?	Identify the domains covered by the ontologies and verify whether they address aspects related to AUI systems (e.g., UI elements) or application domain aspects (e.g., entertainment).
RQ11	Have the ontologies reused others?	Determine if reuse has been common. This helps identify if practices to accelerate ontology development have been considered.
RQ12	Have ontologies been developed based on foundational ontologies?	Determine if there has been a concern with using well-founded ontologies to enhance ontology quality.

RQ1 and RQ2 represent typical SMS inquiries that offer a broad overview of the research subject. The remaining questions refer to specific aspects of (i) the use of ontologies to aid in developing AUI systems and (ii) the used ontologies. Concerning (i), the RQs investigate the purposes for using ontologies (RQ3), the software development phases they are applied in (RQ4), how they have been used (RQ5), the UI adaptations they support (RQ6), and the systematic development methods used (RQ7). As for (ii), the RQs investigate the types of ontologies used (RQ8), the languages for their representation (RQ9), the domains they cover (RQ10), the reuse adoption (RQ11), and the use of foundational ontologies (RQ12). The research questions were identified based on previous studies performed in the author's research group, on papers, books, and other materials on the topic studied during the informal literature review that preceded the SMS, and on the authors' knowledge and experience in the area.

The adopted *search string* comprises two groups of terms linked by the operator AND. The first group includes terms related to "Ontology", while the second encompasses terms related to "Adaptive User Interface". To guarantee broader coverage, we did not associate the term "system" with "Adaptive User Interface". Within each group, the OR operator was used to account for synonyms. The following search string was used: (("ontology" OR "ontologies") AND ("adaptive user interface" OR "adaptive interface" OR "adaptive user interface" OR "adaptable user interface" OR

“adaptable interface” OR “interface adaptation”))². For establishing the string, we experimented with various terms, logical operators, and their combinations. The goal was to identify a string that maximized the retrieval of relevant publications while minimizing irrelevant results. We assessed the effectiveness of each string by analyzing the number of publications retrieved and their relevance to our study. Terms that significantly increased the number of non-relevant publications were excluded (e.g., “user interface”). More restrictive strings that missed important studies identified in our initial literature review (we used them as control publications) were also disregarded.

The **sources** used in the study relied on digital libraries that met specific criteria Costa and Murta (2013): (i) permitting the use of logical expressions or a similar mechanism for searches; (ii) supporting full-length searches; (iii) being accessible within the researchers’ institution; and (iv) covering the research area, specifically Computer Science. The selected sources included Scopus, Web of Science, Engineering Village, and ACM. We selected these digital libraries due to their comprehensiveness and based on other secondary studies performed in our research group. For running the search string in each digital library, we adapted it according to the search mechanism rules. The string adopted in each digital library can be found in the study package, available at (Freitas et al., 2024).

Publication selection was performed in four steps. *Preliminary Selection and Cataloging (S1)* involved applying the search string to each digital library, restricting the search to title, abstract, and keyword metadata fields. In *Duplications Removal (S2)*, publications indexed in multiple libraries were identified, and redundancies were eliminated. *Selection of Relevant Publications (S3) – first filter* – entailed analyzing abstracts considering the following inclusion (IC) and exclusion (EC) criteria: (IC1) the paper addresses the use of ontology in the development of AUI systems (the system as a whole or its UI); (EC1) the paper does not address the use of ontology in the development of AUI systems (the system as a whole or its UI); (EC2) the paper lacks an abstract; (EC3) the paper was presented only as an abstract; (EC4) the paper is a secondary study, tertiary study, editorial, summary of keynotes, tutorials, or proceedings of a scientific event; (EC5) the paper is not written in English. Subsequently, the *Selection of Relevant Publications (S4) – second filter* – involved reading and analyzing the full text of the papers selected in S3, considering, in addition to the aforementioned criteria: (EC6) the publication is an older version - outdated - of a study already selected; and (EC7) it was not possible to have access to the full text of the publication. Figure 1 illustrates the steps followed to select publications.

We used the StArt tool³ to support publication selection. To consolidate data, publications returned in the publication selection steps were cataloged and stored in spreadsheets. We defined an id for each publication and recorded the publication title, authors, year, and vehicle of publication. Data from publications returned in S4 were extracted and organized into a data extraction table oriented to the research questions. The spreadsheets produced during the study can be found in the study package available at (Freitas et al., 2024).

The publication selection process was carried out with the participation of all the authors. Initially, the first, second, and third authors performed steps 1 and 2 of the selection process. The first author then selected an initial set of approximately 10% of the papers and performed steps 3 and 4. The second and third authors performed the same steps independently. The results were compared, discussed with the fourth author, and disagreements were discussed until consensus was reached. The first author then performed steps 3 and 4 for a new sample of 40%, which were also evaluated by the second author. Once

²As the search string includes the terms “ontology” and “ontologies”, we selected papers that explicitly state that an “ontology” or “ontologies” were used in AUI system development context. Papers addressing the use of conceptual models not named ontology in the referred paper were not considered in this study.

³http://lapes.dc.ufscar.br/tools/start_tool

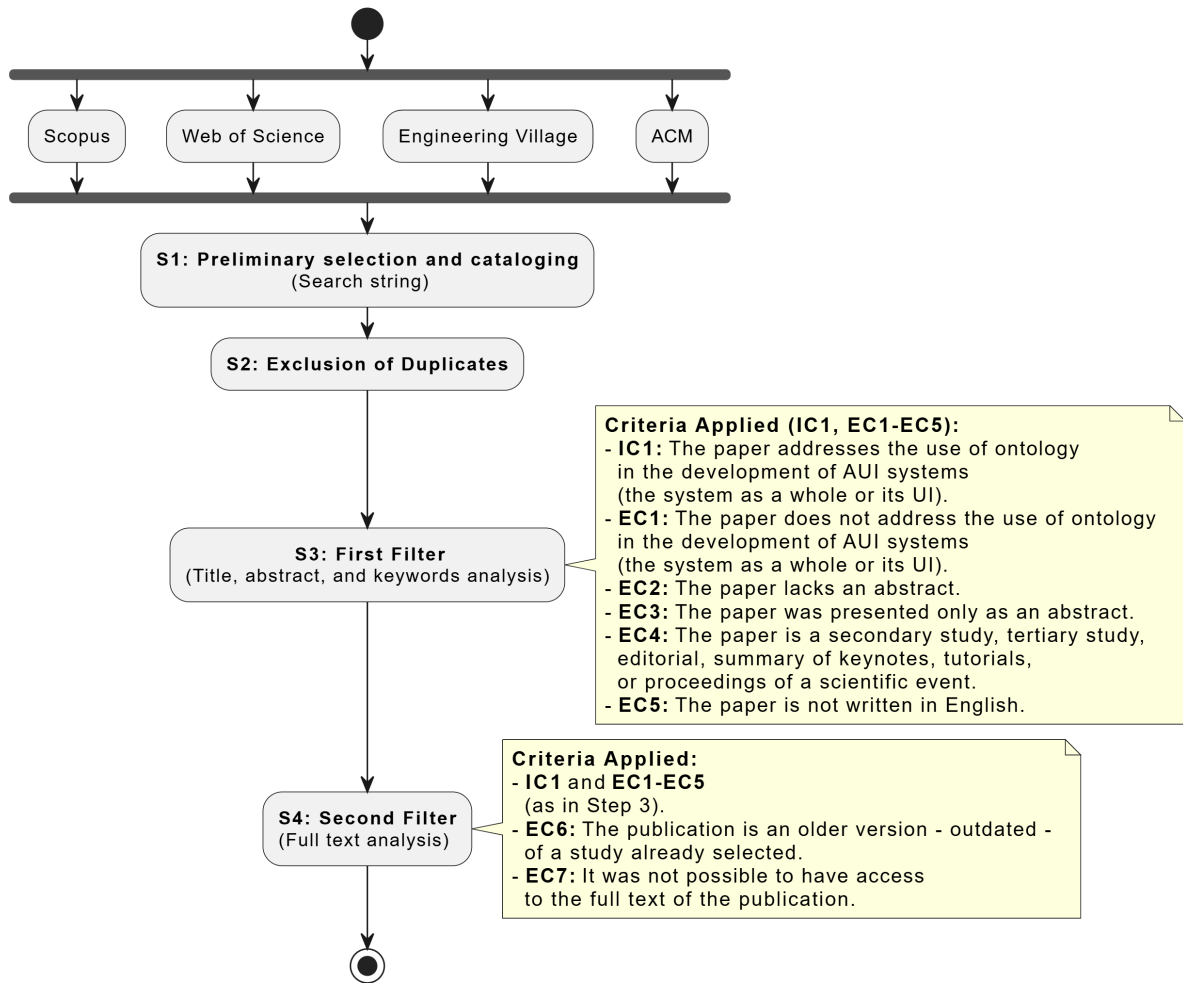


Figure 1. Publication selection process.

again, the results were compared, discussed with the fourth author, and disagreements were resolved. Thus, the first author performed steps 3 and 4 for the remaining papers. All publications selected in step 4 were fully analyzed by the first author and at least one of the other researchers. Data extraction was carried out by the first author, and was reviewed, discussed, and validated with the other authors. Quantitative data were tabulated and utilized in graphs and statistical analysis. Finally, the researchers performed qualitative analysis considering the findings, their relation to the research questions, and the study purpose.

4. Data Extraction and Synthesis

The study encompassed papers published until 2023, with the final searches conducted in March 2024. Figure 2 illustrates the process followed and the number of publications selected at each stage. As

explained earlier, to search the digital libraries, the search string underwent syntactic adaptations to suit the specific characteristics of each source.

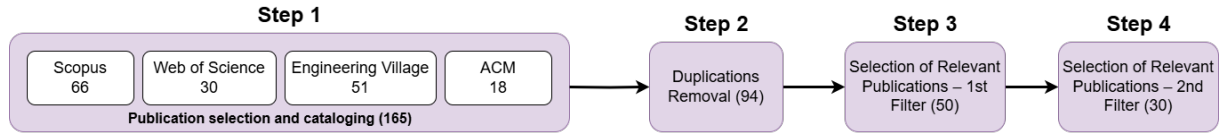


Figure 2. Number of publications selected at each step.

In the first step, a total of 165 papers were retrieved from the selected sources. The second step involved eliminating duplicates, resulting in 94 publications (a reduction of 43%). The third step applied selection criteria to the abstracts, resulting in 50 papers (a 46.81% reduction). At this step, in cases of doubt, papers were selected to be further analyzed in the next step. In the fourth step, selection criteria were applied to the full texts, resulting in 30 publications (a reduction of 40%).

Table 2 provides a brief description of each ontology of the selected publications. From the 30 selected publications, we identified 23 different works because we noticed that some publications addressed complementary works from the same research group. To analyze these cases, we grouped the complementary publications. Therefore, complementary papers by the same authors were grouped into a single data extraction, as follows: two papers into id #01, four papers were grouped into id #03, two papers into id #05, two papers into id #09, and another two papers into id #19. We also must clarify that article #23 refers to an exploratory study performed by this paper's authors before the completion of the SMS. Along this and the following sections, we refer to the papers using the ID listed in Table 2. After, we present the data synthesis for each research question. Detailed information about the selected publications, including a brief description and extracted data, can be found in (Freitas et al., 2024).

Table 2: Selected publications.

ID	Title	Ref.
#01	Ontology on UI design and adaptation patterns in mobile systems.	Braham et al. (2021, 2022)
#02	Accessibility ontology for people with color vision deficiency (CVD).	Bonacin et al. (2022)
#03	Addresses user profile, disabilities, UI components, and devices.	Kultsova et al. (2016, 2017, 2018, 2019)
#04	Focuses on Environment, Device, User Profile, Service, and Adaptive UI for smart parking management.	Ghannem et al. (2016)
#05	Addresses context of use, user profile, device, and environmental factors for UI adjustments.	Soui et al. (2017); Lebib et al. (2016)
#06	Ontology to adapt navigation interfaces on online retail websites, including concepts such as navigation module, page area, priority, user roles, and personal characteristics.	Liu and Huang (2016)
#07	Represents adaptive UIs, user characteristics, context, and devices.	Castillejo et al. (2014b)

ID	Title	Ref.
#08	Addresses users with disabilities, considering context and device settings.	Zouhaier et al. (2013)
#09	Represents web accessibility, user characteristics, interaction contexts, and device parameters.	Zakraoui and Zagler (2011, 2012)
#10	Represents user, device, and environment.	Kong et al. (2011)
#11	Describes user affective interactions (information about the user's sensory and perceptual capabilities) and system characteristics.	Cearreta and Garay-Vitoria (2011)
#12	Addresses the user profile, including visual acuity, color blindness, hearing impairment, and user context.	Gacimartín et al. (2011)
#13	Describes user, device, and physical environment.	Hervás and Bravo (2011)
#14	Addresses the skills and characteristics of users, device parameters, and the characteristics of services or targets (such as ATMs and vending machines).	Miñón et al. (2010)
#15	Represents user profile, context profile, and user interaction.	Partarakis et al. (2009)
#16	Models an environment and its graphical UI in the field of ambient intelligence.	Gómez et al. (2008)
#17	Represents UI components and their relationships.	Vermeulen et al. (2007)
#18	Used existing ontologies to organize types of visual UI presentations and communication between the user and the system.	Crapo et al. (2003)
#19	Represents context information, including user context, device context, environment context, and activity context.	Iqbal et al. (2021, 2022)
#20	Represents accessibility concepts such as the user disability profile, the user situations, the user profile, the technical context, the environment context, and IoT.	Souidi et al. (2022)
#21	Represents disability profile, environment context, user situation, and accessibility context.	Zouhaier et al. (2017)
#22	Represents knowledge of contextual factors, including user profile, current task, environment, and device parameters.	Stefanidi et al. (2022)
#23	Represents AUI system, user profile elements, context of use, and UI components.	Freitas et al. (2022)

Publication year and vehicle (RQ1): Figure 3 shows the distribution of the 30 selected publications over the years and the distribution of publications considering the vehicle of publication. Publications addressing the use of ontologies to aid in the development of AUI systems have spanned since 2003. Since then, papers addressing this subject have been published in journals and conferences (no workshop publication was found). *Conferences* have been the main forum, encompassing 73.3% of the publications (22 out of 30). Eight papers (26.7%) were published in *journals*. In Figure 3 we omitted years in which no publication was found. Hence, looking at the x-axis it is possible to notice some intervals where no publication was identified. The more expressive is the 5-year gap between 2003 and 2008.

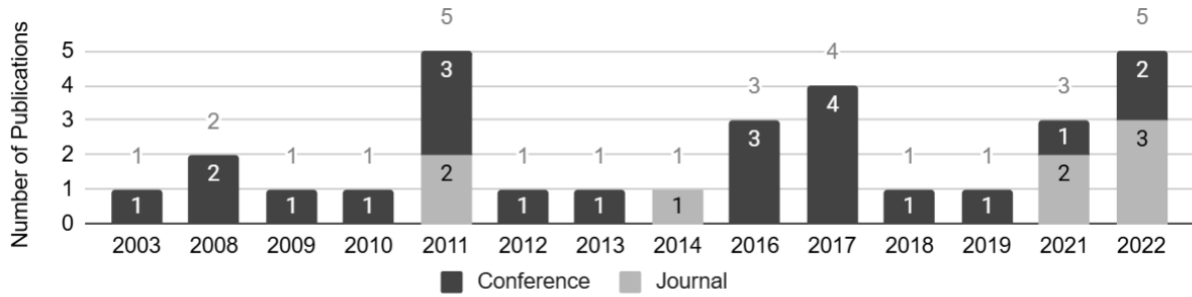


Figure 3. Publications over time.

Research type (RQ2): Considering the classification defined by Wieringa et al. (2006), from the 30 publications, the majority (63.3%) combined *Proposal of Solution* and *Validation Research* (19 papers). An additional 23.3% (seven papers) focus solely on *Proposal of Solution*. A smaller portion (6.7%) combined *Evaluation Research* and *Proposal of Solution* (two articles). Additionally, another 6.7% included a combination of *Evaluation Research*, *Proposal of Solution*, and *Validation Research* (two articles). Figure 4 provides a visual breakdown of the research types.

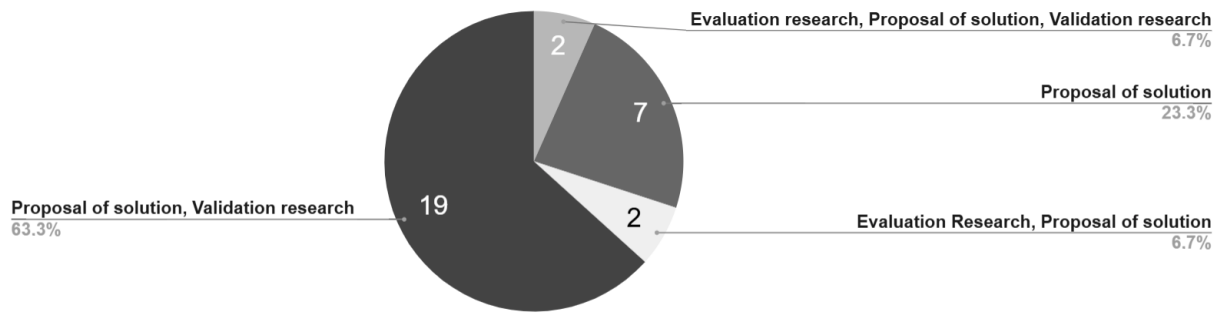


Figure 4. Research type.

Purposes for using ontologies (RQ3): In this RQ, we considered the categorization proposed in Costa et al. (2021). Some ontologies were classified into more than one category. Figure 5 illustrates the results.

Being a *Basis of a System/Approach* means that the ontology is incorporated as part (for example, a module or a repository) of the architecture of the proposed system or approach. For example, an adaptive e-learning system might use an ontology as a module to define the characteristics of students with disabilities and as a repository to store rules that adapt the visual and interactive elements. Ontologies used for *Conceptual Description* aim to provide a formal specification of a portion of the domain of interest (e.g., defining and organizing fundamental concepts of the AUI system). *Semantic Annotation* serves to semantically annotate data and content (e.g., instructions or conditions that determine how a system should behave or adjust in response to different situations or user characteristics), allowing machines to read it. Semantic annotations identify and categorize relevant data, associating them with ontological concepts.

All 23 ontologies (100%) were used as a *Conceptual Description*. For example, #01 is part of a framework that is based on three modules: the Design Pattern module, the User Profile module, and

the User Interface module. The ontology models and describes aspects of the domain, such as interface characteristics, user profiles, and usage context. Similarly, in #10, the approach reuses existing ontologies to describe interaction context, detailing the state in which a person uses a device and UI characteristics.

22 (95.7%) ontologies (#01, #02, #03, #04, #05, #06, #07, #08, #09, #11, #12, #13, #14, #15, #16, #17, #18, #19, #20, #21, #22, and #23) were used as a *Basis of System or Approach*. For example, #03 is used as the *basis of* adaptive architecture, integrating domains such as users, diseases, devices, and interface patterns to support rule-based reasoning. It *describes concepts* of interaction problems, contexts, and design patterns.

Nine (39.1%) ontologies (#03, #04, #06, #09, #11, #16, #17, #22, and #23) were used as a *Semantic Annotation*. In #03, it uses *semantic annotations* to change interface elements, allowing dynamic customization of properties such as font size and multimedia controls.

There was a single paper (#10) in which the ontology was used only for *Conceptual Description*. In all the other papers, the ontology was used for multiple purposes. In 13 papers (56.5%) (#01, #02, #05, #07, #08, #12, #13, #14, #15, #18, #19, #20, and #21) ontologies were used as a *Basis of System/Approach* and to aid *Conceptual Description*. In nine papers (39.1%), ontologies (#03, #04, #06, #09, #11, #16, #17, #22, and #23) were used as a *Basis of System/Approach*, *Conceptual Description*, and *Semantic Annotation*.

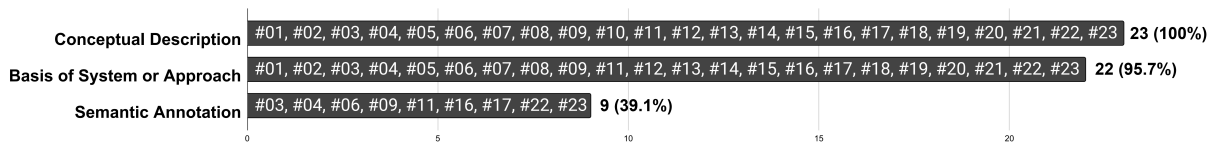


Figure 5. Purpose for using ontologies.

System development phase in which ontologies have been used (RQ4): This RQ identifies the phases in which ontologies have been employed (e.g., *Requirements Specification* (identifying and documenting stakeholder needs and expectations to guide development) Sommerville (2020), *Modeling* (creation of abstract representations of the system, such as diagrams, to understand and plan its structure and behavior) Booch (2005), *Design* (detailed planning of how the system will be built, specifying components and interfaces) Gamma (1995), *Implementation* (transformation of design into functional source code using programming languages) McConnell (2004), and *Testing* (verification and validation process to ensure that software meets requirements and is free from fault) Myers et al. (2011)).

Among the 23 works analyzed, 15 (65.2%) applied ontologies in the *Implementation* phase. For example, in #23, the reference ontologies were translated into operational formats and integrated into a reasoning engine to enable runtime adaptation of the UI. Additionally, 10 (43.5%) papers reported using ontologies during the *Requirements Specification* phase. For example, #23 used ontologies to help identify user characteristics, aligning domain understanding and ensuring completeness and consistency in requirements. Ontologies were also employed in the *Design* phase in 10 (43.5%) papers. For example, in #23, the ontologies were incorporated into the system architecture, defining the relationships and interactions between components. Ontologies were utilized in the *Modeling* phase in seven papers (30.4%). For example, #23 used the ontology model to create the class diagram and define the database structure. Lastly, none of the papers applied ontologies in the *Testing* phase.

How ontologies have been used to support the development of AUI systems (RQ5): In this RQ, the categorization was established based on the data extracted from the articles. The following categories were considered: *Reference Models for the System Structure*: the ontology is used as a reference model to define

Table 3
Development phase in which ontologies have been used.

ID	Requirements Specification	Modeling	Design	Implementation	Testing
#01			✓	✓	
#02	✓	✓	✓	✓	
#03				✓	
#04				✓	
#05	✓				
#06	✓				
#07	✓				
#08	✓				
#09				✓	
#10				✓	
#11				✓	
#12			✓	✓	
#13				✓	
#14				✓	
#15	✓				
#16			✓	✓	
#17		✓	✓	✓	
#18			✓	✓	
#19	✓	✓	✓		
#20		✓	✓		
#21		✓			
#22	✓	✓	✓	✓	
#23	✓	✓	✓	✓	

the system structure and relationships between system components Cabac et al. (2014); *Source of Design Patterns of UI Adaptation*: the ontology is used as a source of design patterns of solutions for common UI adaptation needs, such as font size adjustments or layout reorganization Braham et al. (2022); *Knowledge Integration Artifact*: the ontology is used as a structure for developing knowledge bases O’Leary (1998); *Semantic Reasoning for UI Adaptation*: the ontology is used in inference engines to adapt systems Petnga and Austin (2016). Some works were classified into more than one category. For example, #03 uses ontology to define the system structure and relationships between system components (*Reference Models for the System Structure*). The ontology was also used to model and structure the domains of user characteristics, diseases, interfaces, and devices, integrating them into a unified meta-ontology for AUI (*Knowledge Integration Artifact*). It provided a foundation for the development and application of UI

design patterns (*Source of Design Patterns of UI Adaptation*). Finally, the ontology was employed in reasoning mechanisms using semantic rules (e.g., SWRL) to infer appropriate adaptations and generate personalized UI profiles for users with special needs (*Semantic Reasoning for UI Adaptation*). Table 4 illustrates the results.

Table 4
How ontologies have been used.

ID	Reference Models for the System Structure	Source of Design Patterns of UI Adaptation	Knowledge Integration Artifact	Semantic Reasoning for UI Adaptation
#01		✓	✓	✓
#02		✓	✓	✓
#03	✓	✓	✓	✓
#04	✓		✓	✓
#05				✓
#06	✓			
#07	✓		✓	✓
#08	✓			
#09	✓			
#10	✓			
#11	✓			
#12	✓			
#13	✓	✓	✓	✓
#14	✓			
#15	✓			
#16	✓			✓
#17	✓			✓
#18	✓			✓
#19	✓		✓	✓
#20	✓			✓
#21	✓		✓	
#22	✓			✓
#23	✓	✓	✓	✓

20 papers (87%) of the 23 papers used ontologies as *Reference Models for the System Structure*. For example, #13 used ontologies to model contextual information and define the structural elements of the system, while #23 used to map adaptive computer systems. 14 papers (60.9%) used ontologies for *Semantic Reasoning for UI Adaptation*. For example, #13 the ontology has been employed in semantic reasoning engines, using rules (e.g., SWRL) to infer appropriate adaptations and generate adapted UI. #23 used the ontology to infer appropriate UI adaptations. 39.1% (nine out of 23) used ontologies as *Knowledge Integration Artifact*. For example, #13 used the ontology to model and structure domains such as user characteristics, devices, and environmental contexts. #23 used the ontology to integrate knowledge in some subdomains, including user, user disabilities, AUI system, UI elements, and characteristics.

Finally, five papers (21.7%) used ontologies as *Source of Design Patterns of UI Adaptation*. For example, in paper #13, ontologies were employed to generate visualization mosaics dynamically, applying adaptive design patterns such as “news panels” for general content display and “document viewers” for focused interactions. #23 defines reusable and structured rules that guide UI adaptations in some scenarios, such as adjusting font size for users with low vision or enabling dark mode for people sensitive to screen brightness.

In general, the ontologies are used in the analyzed papers to address complex aspects of AUI systems development. The motivation for that lies in the ability of ontologies to organize domain knowledge explicitly (e.g., #01, #03, #07, #13, #16, #17, #23) and, more importantly, to serve as a basis for reasoning that enables UI adaptation at runtime (e.g., #03, #05, #13, #22, #23). Ontologies help infer new information from contextual data through inference engines, especially when they are expressed in languages such as OWL (e.g., #01, #03, #22). Another motivation for using ontologies regards their ability to facilitate knowledge reusability and interoperability, enabling the integration of different data sources and the extension of existing models (e.g., #03, #07, #09, #11, #13).

UI adaptation supported by ontologies (RQ6): This research question explores the varied applications of ontologies in supporting UI adaptation. As depicted in Figure 6, 22 (95.7%) of the 23 papers reported the use of ontologies for adapting *UI Elements*, encompassing visual interface components such as buttons, icons, menus, and other elements the UI. 13 (56.5%) papers detailed the utilization of ontologies to adapt *Contents* (customization and transformation of the content displayed in the interface (e.g., adding subtitles or transcribing audio into text)) Iqbal et al. (2022). 12 (52.2%) papers employed ontologies for *Interactions* of users with the AUI system, encompassing output modes like audio, video, ringtone, vibration, graphics, text, and voice notification. Finally, three (13%) papers applied ontologies to enhance user *Navigation* on the system, facilitating more streamlined access to required information.

Some papers fall into more than one category. For example, in paper #06, visual components, such as checkout modules and menus (*UI Elements*), are dynamically adapted based on roles defined by the ontology. In the navigation structure (*Navigation*), such as “one page” or “step by step” checkout processes, it is configured according to the navigation module ontology. Paper #07 uses ontologies to support user UI adaptations, covering visual elements (*UI Elements*), content (*Contents*), interactions (*Interactions*), and navigation (*Navigation*).

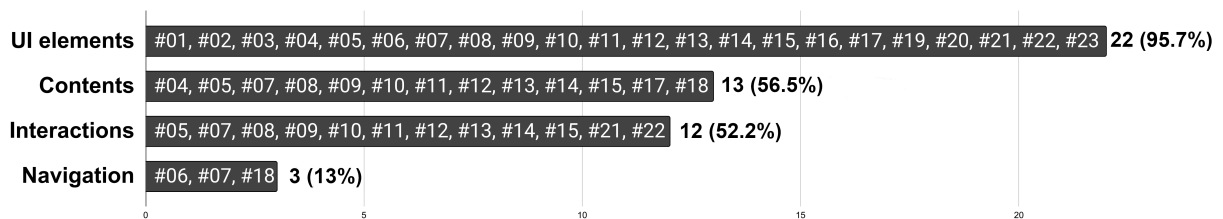


Figure 6. UI adaptations supported by ontologies.

Use of systematic method (RQ7): To answer RQ7, we first investigated how many of the 23 papers followed a systematic method to develop the system (i.e., a set of steps guiding the system development). As illustrated in Figure 7, eight (34.8%) papers used a systematic approach to develop the AUI systems, while 15 (65.2%) did not.

We then analyzed the eight papers that used a systematic approach to develop the system to identify whether the adopted systematic approaches addressed the use of ontologies to develop the system (i.e.,

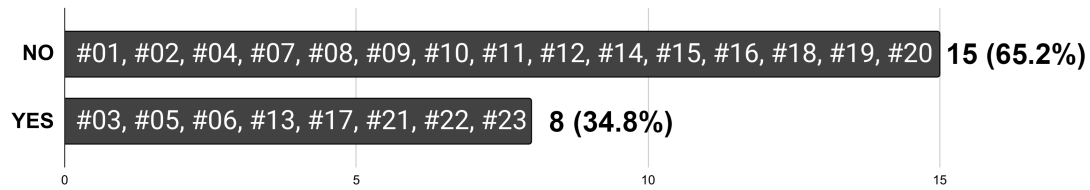


Figure 7. Use of a systematic approach to develop the system.

they provide some guidance to use ontologies when developing the system). From the eight papers, seven (#03, #05, #13, #17, #21, #22, and #23) (87.5% of the eight papers) used systematic approaches that address the use of ontologies in the development of the system. Next, we briefly describe these seven approaches. Figure 8 illustrates the results regarding the adoption of systematic approaches that addressed the use of ontologies to develop the system.

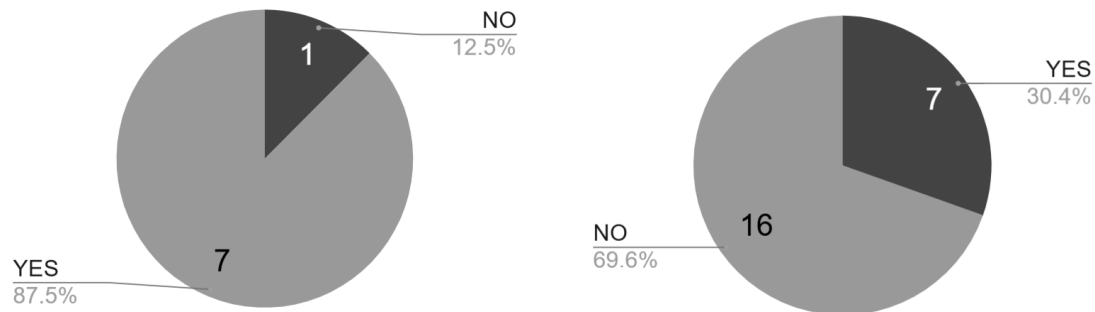


Figure 8. Guidance on the ontology use in the systematic method to develop the system: (a) considering only papers that adopted a systematic approach; (b) considering all the papers.

In #03, a two-phase process for developing AUI systems is presented. The first phase includes two procedures executed by the software developer: (1) the creation of an interface mock-up; (2) the adaptation of the interface according to the user context, considering the user's disease, peculiarities caused by the condition, the device used, and other factors. The second phase involves two procedures: (1) the evaluation of the application by users and their caregivers during interaction with the interface; (2) the personalization of the interface based on information about user behavior and possible changes in their context. The first phase of the process is based on inference from expert knowledge represented in a domain ontology, while the second phase utilizes interface patterns represented in the pattern ontology.

#05 presents a structured method for developing AUI based on ontologies, divided into four main steps: (1) correlation of context criteria (e.g., age, motivation) and interface characteristics (e.g., density, guidance), identifying their relationships; (2) ontological modeling, creating an ontology to formalize the adaptation process; (3) ontological reasoning, using inference engines to generate adapted UIs based on predefined rules; and (4) evaluation of the proposed UIs, detecting issues and refining the process. The method focuses on the development of AUI, without addressing the system as a whole.

#13 followed a systematic method that includes five steps: (1) Modeling contextual knowledge, where ontological information is used to structure data about the user, the environment, and the available devices, allowing a simplified representation of the world around them; (2) Each user interacts with the system,

providing information that is integrated into the context model, including personal characteristics and preferences; (3) Automatic generation of user interfaces, which dynamically adjust in real time based on contextual information and user needs, using the ViMos framework infrastructure; (4) Evaluation of system effectiveness, where user feedback is collected to measure satisfaction with the adapted interfaces and the relevance of the information presented; and (5) Modeling personalization rules, which allows the continuous adaptation of interfaces, ensuring that users receive relevant information according to their specific needs and contexts.

#17 followed a systematic method consisting of five steps: (1) Collecting the required services using the OWL-S ontology to retrieve, rate, and calculate the average rating of pictures; (2) Creating a hierarchical task model, enabling structured user interactions; (3) Designing or reusing layout templates by defining spatial relationships and hierarchical layout elements; (4) Instantiating the layout templates and adding additional interface elements; and (5) Rendering the UI.

#21 proposed a generic adaptation process based on the Model Driven Engineering (MDE) methodology for the adaptation of UIs considering the accessibility context divided into four main steps: (1) Abstraction of the non-adapted interface into a Platform Independent Model (PIM) using reverse engineering, eliminating implementation details and modalities; (2) Development of an ontology that describes the accessibility context, including disability profiles, user situations, and technical and environmental context, serving as input for the adaptation process; (3) Adaptation process, which transforms the non-adapted PIM meta-model into a disability-specific meta-model (Disability-Meta-model), parameterized by the accessibility context ontology; and (4) Specific transformation of the adapted PIM into a Platform Specific Model (PSM), adjusted to the modalities supported by the execution platform.

In #22, the AUI system was developed using a methodology to adapt the UI in real-time, focusing on the user's situational awareness (SA). The core of this approach is the Decision Maker (DM) module, which determines what information to display, how to present it, and where to position it in the UI. The methodology includes: (1) Context module, which extracts the relevant context information regarding the user's profile (e.g. expertise) and state (e.g. stress level), the environment (e.g. crowdedness), and the task/activity at run-time (e.g. incident resolution); (2) Knowledge Base Module, which provides to the DM the necessary data (Information Elements) regarding the current scene (e.g. detected humans, identity information) and human input (e.g. feedback from the Command & Control); and (3) Visualizer Module, which contains the collection of design 'templates' for all the supported GUI elements and performs the rendering in the display. The DM integrates three interconnected units: (i) Ontology Model, which captures and dynamically updates GUI metadata and context factors; (ii) SA Reasoner, which assigns SA scores to GUI elements based on ontology rules and current context; and (iii) UI Optimizer, which uses these scores and constraints to compute the optimal GUI layout for real-time adaptation.

Finally, #23 development of the AUI system in the study followed a systematic ontology-based method following of seven steps: (1) Identify System Scope and Users, defining the purpose, boundaries, and user types, with a focus on characteristics that will guide interface adaptations; (2) Elicit System Requirements, specifying functional and non-functional requirements, including specific UI adaptation needs; (3) Select Ontology, choosing a relevant fragment from an ontology network (ON) aligned with the defined requirements; (4) Perform System Analysis, developing structural and behavioral models based on the reference ontology; (5) Define System Architecture, defining the system architecture, its components (e.g., problem domain component, UI component) and related technologies; (7) Develop Operational Ontology, transforming the reference ontology into a machine-readable format, incorporating adaptation rules verified by reasoning tools; and (8) Implement and Test the System, ensuring that the system meets the established requirements and that the UI is functional and adaptive.

Type of ontology used (RQ8): 13 (56.5%) publications employed *operational ontologies*, nine (39.1%) used both *Operational* and *Reference* ontologies, and one (4.3%) did not inform which type of ontology was used. Figure 9 details the distribution of articles according to the type of ontology.

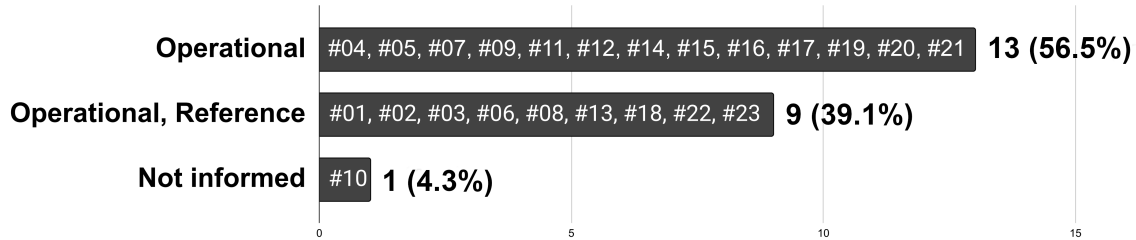


Figure 9. Ontology type.

Languages used for ontology implementation/representation (RQ9): Considering that reference ontologies and operational ontologies often use different languages, we separated the results per ontology type.

Concerning *operational ontologies*, six (25%) papers did not specify the language used. 11 papers (45.8%) used OWL (Ontology Web Language), while three papers (12.5%) employed OWL-DL, a sublanguage of OWL that combines maximum expressiveness with computational completeness and decidability, known for its heightened expressivity and reasoning capabilities compared to OWL. Two papers (8.3%) employed OWL2, an extension of OWL1, with improvements. Finally, one paper (4.2%) utilized OWL-S, which describes services and their interactions. Additionally, one paper (#15) mentions more than one language (OWL and OWL-DL) for their ontologies. Figure 10 shows results related to operational ontologies.

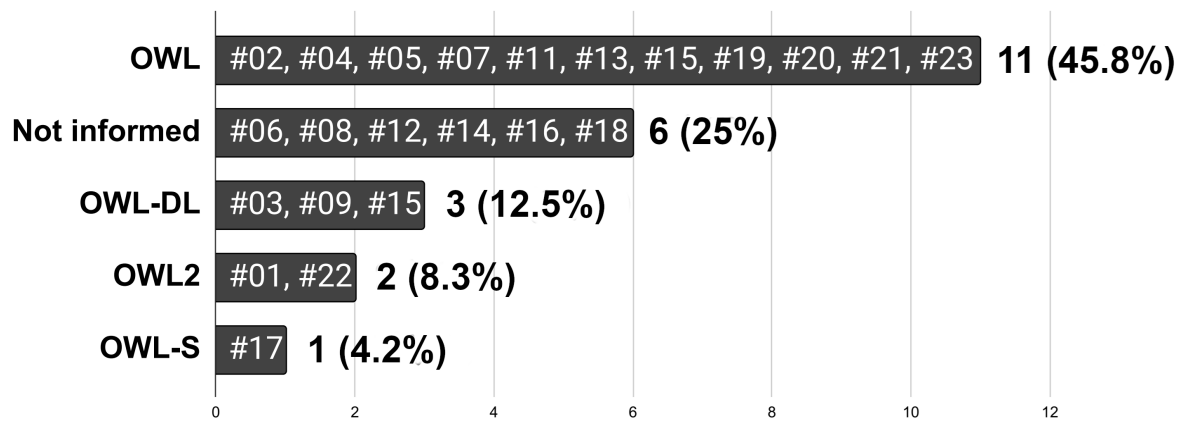


Figure 10. Implementation and representation languages used in operational ontologies.

Concerning reference ontologies, as illustrated in Figure 11, seven (30.4%) of the 23 papers did not report the language used to develop their ontologies. One (4.3%) paper (#03) employed the use of IDEF5 Diagram, a software engineering method used to develop and maintain usable and accurate domain

ontologies. Furthermore, another (4.3%) paper (#23) explicitly stated the use OntoUML⁴, an extension of the UML (Unified Modeling Language) designed to build well-founded conceptual models.

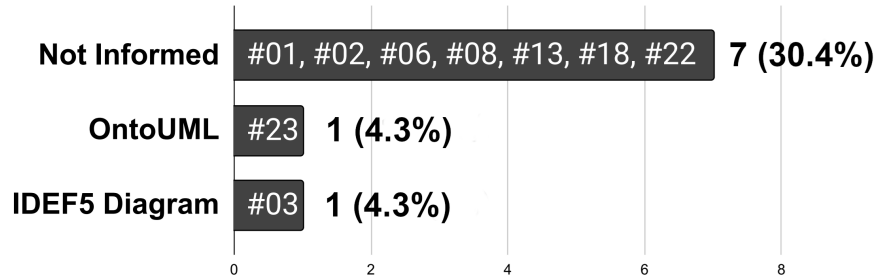


Figure 11. Implementation and representation languages used in reference ontologies.

Domains covered by the ontologies (RQ10): Table 5 shows the domains addressed by the ontologies identified in the study. The categorization used to classify the domains was established based on the extracted data, namely:

- *Context of Use*: Addresses conditions and scenarios in which the application or system can be used, such as the physical or social environment Ramaprasad and Papagari (2009).
- *Device*: Devices used by users, such as smartphones, tablets, and desktops, which influence interaction with the UI Iqbal et al. (2022).
- *User Profile*: Information about user characteristics, such as age, gender, and technological experience Chebba et al. (2015).
- *User Disability*: Physical, sensory, or cognitive limitations of users Bravo et al. (2019).
- *UI Elements*: Represents the visible and interactive components of the user interface, such as buttons and menus Feng et al. (2006).
- *Environment*: External factors that influence interaction, such as lighting, noise, or connectivity Jaskolka et al. (2015).
- *User Interaction*: Involves monitoring and analyzing how users interact with the system (e.g., through voice commands, gestures, clicks, or other interactions) Kultsova et al. (2017).
- *Adaptive UI*: Customization rules that allow the UI to be adapted Castillejo et al. (2014b).
- *User*: Refers to the individual using the interface, considering their needs and objectives Bouquet et al. (2004).
- *User Preferences*: Users' personal choices, such as interaction style, colors, and content organization Loitsch et al. (2017).
- *UI Design Patterns*: Reusable solutions to common interface design problems, such as navigation and layout Braham et al. (2021).

The domain most frequently addressed by AUI ontologies is the *Context of Use*, with 16 works covering this domain (69.6%). 13 (56.5%) ontologies focus on the *Device*, 13 (56.5%) on *User Profile*, 10 (43.5%) on *User Disability*, 10 (43.5%) on *UI Elements*, nine (39.1%) on *Environment*, nine (39.1%) on *User Interaction*, eight (34.8%) on *Adaptive UI*, six (26.1%) on *User*, three (13%) on *User Preferences* and two (8.7%) on *UI Design Patterns*. Table 5 provides a view of the papers and the respective domains they cover.

⁴<https://ontouml.org/>

Table 5
Covered domain of the ontologies.

Domain	Papers	Total	Percentage
Context of Use	#01, #02, #04, #05, #07, #09, #12, #13, #14, #15, #16, #18, #19, #20, #21, #22	16	69.6%
Device	#02, #03, #04, #05, #07, #10, #13, #14, #16, #19, #20, #21, #22	13	56.5%
User Profile	#01, #04, #05, #06, #07, #12, #13, #14, #15, #20, #21, #22, #23	13	56.5%
User Disability	#01, #02, #03, #08, #09, #12, #15, #20, #21, #23	10	43.5%
UI Elements	#01, #02, #03, #05, #06, #11, #16, #17, #18, #23	10	43.5%
Environment	#04, #05, #10, #13, #16, #19, #20, #21, #22	9	39.1%
User Interaction	#01, #02, #05, #06, #09, #10, #11, #15, #22	9	39.1%
Adaptive UI	#01, #02, #03, #04, #05, #11, #13, #23	8	34.8%
User	#01, #03, #10, #13, #22, #23	6	26.1%
User Preferences	#01, #02, #21	3	13%
UI Design Patterns	#01, #03	2	8.7%

Ontology Reuse (RQ11): In 14 (60.9%) papers, the ontologies used did not reuse concepts from existing ontologies. In the remaining 9 (39.1%) papers, the ontology employed reused concepts from other ontologies. For example, #03 used the device model of the AEGIS Ontology⁵ project to include concepts about hardware resources and assistive software. #04 used a model proposed in previous work on environment Glouberman et al. (2006) to include the dimensions of Environment, Device, User, and Service. In #07 and #09, the FOAF⁶ ontology was used to complement the user profile. #11 reuses an existing ontology, and the emphasis is on choosing appropriate features to describe interaction context according to user tasks (there is no information about the name of the ontology used). #13 extended FOAF to characterize cognitive aspects of users. #12 extended the Affinto ontology Cearreta Urbieto (2010) to include information about users' sensory and perceptual capabilities. #17 used OWL-S⁷, an OWL-based web service ontology. Finally, #23 utilized ontologies from an ontology network Costa et al. (2020), focused on Human-Computer Interaction subdomains. Figure 12 illustrates these results.

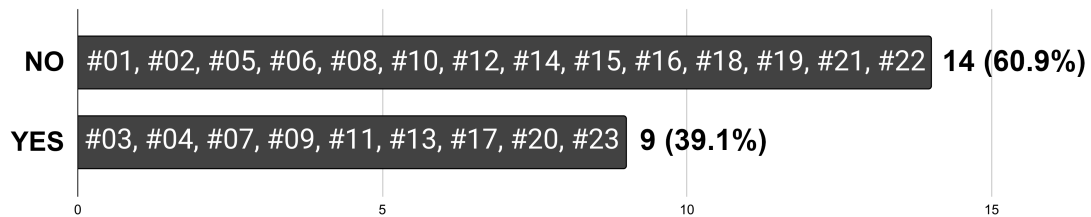


Figure 12. Ontology reuse.

Have ontologies been developed based on foundational ontologies? (RQ12): According to the publications, only one (4.3%) ontology (#23) has been developed based on foundational ontologies, utilizing UFO (Unified Foundational Ontology) (Guizzardi, 2005).

⁵"AEGIS Ontology provides support for the formal and unambiguous definition of accessibility domains, and the possible semantic interactions between them." Torres-Carazo et al. (2021)

⁶<http://xmlns.com/foaf/spec/>

⁷<https://www.w3.org/submissions/OWL-S/>

5. Data Analysis and Discussion

In this section, we discuss the results obtained based on the RQs.

Regarding *when and where* (RQ1), the papers have been published and the topic has been investigated for over 21 years, indicating that the subject is not new. However, a significant gap between 2003 and 2008 was observed, during which no publications were identified. Although research on this topic has been frequent in recent years, the big picture suggests that the topic can be further explored. There has been a predominance of conference papers (73.3%), which may reflect the evolving and exploratory nature of the topic.

Regarding the *research type* (RQ2), the majority of the papers (63.3%) combined *Proposal of Solution* and *Validation Research*, demonstrating a focus on introducing new proposals and verifying their applicability. Additionally, 23.3% of the papers presented only *Proposal of Solution*, indicating exploratory contributions. Smaller percentages included *Evaluation Research*, with only two papers combining it with *Proposal of Solution* (6.7%), and two combining it with both *Proposal of Solution* and *Validation Research* (6.7%). These findings show that while the field has a significant focus on proposing and validating methods, there is a gap in extensive evaluation studies, suggesting that the proposed approaches have not reached practical settings (e.g., the software industry).

As explained before, our study aimed to investigate (i) the use of ontologies to aid in developing AUI systems and (ii) the used ontologies. Therefore, in the following, we discuss the results by grouping the RQs related to each of these aspects.

Concerning the (i) *use of ontologies to aid in developing AUI systems*, we noticed that in *purposes for using ontologies* (RQ3), the predominance of ontologies used as *System or Approach Base* and *Conceptual Description* highlights the ability to provide a conceptual infrastructure, organizing the main elements of the AUI systems domain, such as user profiles, devices, UI characteristics and context of use. This integration facilitates not only the implementation of the systems but also the alignment between the necessary adaptations and the architecture of the system that will be developed. In addition, *Semantic Annotation* (39.1%) makes it possible to automate adaptive processes, such as dynamic UI adjustments based on the context of use, devices, or user characteristics. This combined use illustrates how ontologies can link conceptual modeling and the implementation of AUI systems.

Regarding the *development phase where ontologies were used* (RQ4), the *Implementation* phase is the most frequent (more than 65%) in the use of ontologies. The ontologies are used to operationalize concepts and functionalities, such as runtime adaptation. This result highlights the role of ontologies as facilitators in the transformation of abstract concepts into executable components, allowing adaptation rules and semantic reasoning to be applied directly in the system to support dynamic UI adaptations. Operational ontologies, being machine-interpretable, are particularly suitable for this purpose as they are designed to ensure desirable computational properties and support application logic at runtime. Ontologies have also been used in conceptual phases, such as *Requirements Specification* (43.5%), *Design* (43.5%), and *Modeling* (30.4%), where they contribute to structuring requirements, guiding architectural planning, and supporting the modeling of system components. No uses were identified in the *Testing* phase. This may be a potential field for future research, especially in AUI systems that demand high quality and reliability.

The predominance of the use of ontologies in the implementation phase is directly related to the predominance of operational ontologies (shown in RQ8). Moreover, in systems development in general, there is a tendency to concentrate efforts on the most tangible phases, especially implementation, where results are more immediate and observable. Operational ontologies, being implemented and directly

integrable into systems, allow for the computational representation of domain knowledge and adaptation rules. Thus, they provide direct benefits to system execution, such as support for inference, dynamic personalization, and interoperability among components.

In contrast, reference ontologies require a higher level of abstraction, aimed at the conceptual representation of the domain and the definition of shared vocabularies. Although these ontologies offer important advantages (such as promoting knowledge reuse, semantic consistency, traceability across development phases, and use in model-driven development), they may demand greater investment in analysis and conceptualization activities, as well as a more systematic integration with software engineering processes. Moreover, there is still a lack of mature approaches that guide the incorporation of ontologies in the early stages of the lifecycle.

Another relevant factor that can influence the use of ontologies is that the available tools and frameworks generally provide direct support for the use of operational ontologies (e.g., through languages such as OWL and inference engines) and lack support for integrating reference ontologies into requirements engineering or conceptual analysis activities. We believe that this methodological and technological gap has reinforced the adoption of ontologies in the implementation phase of AUI systems development.

Regarding *how ontologies have been used* (RQ5), the predominance of ontologies (20 out of 23) as *Reference Models for System Structure* (87%) highlights their fundamental role in defining and organizing AUI system components. Furthermore, their operational value is evident in the use of *Semantic Reasoning for UI Adaptation* (14 out of 23) (around 70% of the articles). Although fewer articles (nine out of 23) employed ontologies as a *Knowledge Integration Artifacts* about AUI systems (39.1%) or as a *Source of Design Patterns of UI Adaptation* (five out of 23) (around 22% of the articles), these roles are nevertheless significant, reflecting the potential of ontologies to unify knowledge and provide reusable design solutions. It is observed that most of the analyzed ontologies prioritize defining specific rules for UI adaptation directly in the system, rather than incorporating them in the ontology itself.

Looking at the *UI adaptations supported by ontologies* (RQ6), ontologies have been used to adapt different aspects of user interfaces. The most frequent category is the adaptation of *UI Elements* (around 96% of the papers), covering visual components such as buttons, icons, menus, and other elements directly perceptible by users. Next, adaptations of *Contents* (56.5%), related to the customization and transformation of the displayed information, such as adding subtitles or transcribing audio into text, and *Interactions* (52.2%), involving adjustments to output modes and the ways users interact with the adaptive system. These adjustments include audio, video, text, and voice notifications, as well as vibration and graphics, allowing the interface to respond in a multimodal (e.g., the UI can present information or enable interactions through a variety of formats, such as audio, video, text, voice, graphics, and vibration). The predominance of these aspects indicates that ontologies have been primarily applied to enhance the adaptability of visual and informational components of UI, as well as to support multimodal interactions.

When analyzing the *use of systematic methods* (RQ7), we noticed that approximately 65% of the papers did not follow well-defined steps for the development of AUI systems. Only eight studies used a systematic approach, and seven of them (30.4% of the total studies) guided the use of ontologies in system development. For example, #03 proposes a two-phase process, combining prototyping and adaptation based on domain ontologies with continuous evaluation and customization. #05 uses a four-step approach, involving the correlation of contextual criteria, ontological modeling, rule-based reasoning, and iterative evaluation of UI. #13 follows a five-step method, with an emphasis on context modeling, integration of dynamic information, real-time adaptation, and continuous system evolution. Finally, #23 presents a comprehensive methodology, incorporating the selection of ontological network fragments, definition of architectures, and transformation of ontologies into operational formats to support adaptation rules. The

results highlight that, although systematic methods are not yet widely adopted in the development of AUI systems, the integration of ontologies in these methodologies offers significant advantages. Ontologies provide a structured basis for capturing and reasoning about information related to the context and users. This analysis reinforces the relevance of ontologies as facilitators of systematic practices by offering a solid basis for the development of AUI systems.

When analyzing the (ii) **used ontologies**, concerning their *types* (RQ8), we noticed that *operational ontologies* have been prevalent (around 57% of the papers). Although 39% of the papers used *reference ontologies*, the use is combined with operational ontologies. Notably, none of the articles adopted just the *reference ontologies*. The predominance of operational ontologies suggests an emphasis on computational aspects when using ontologies to aid in developing AUI systems. The absence of works that exclusively used reference ontologies indicates that although they play an important role in the conceptual definition of the domain and system structure, ontology-based development of AUI systems has relied mainly on operational concerns.

Looking at the *languages* used to represent or implement the ontologies (RQ9), in the papers that used *operational ontology*, there has been a preference for OWL (45.8%), a well-established language for ontology representation in the *Semantic Web*. It is noteworthy that 25% of the papers did not specify the language used for operational ontologies. Concerning *reference ontologies*, of the nine (39%) articles that used reference ontology, only two (8.6%) used established language: IDEF5 Diagram, a structured engineering method for ontology design (#03), and OntoUML, an ontologically well-founded language for ontology-based conceptual modeling (#23). This suggests a lack of a standardized language for representing reference ontologies in the studied context. While operational ontologies rely on standardized languages such as OWL to facilitate computational reasoning and interoperability, reference ontologies appear to use more varied notations to achieve their conceptual modeling goals.

Concerning the *domain covered* (RQ10), we observed that Context of Use (69.6%), Device (56.5%), and User Profile (56.5%) were the domains most covered by ontologies related to AUI systems. The Adaptive UI (34.8%) and UI Design Patterns (8.7%) domains were little explored by the identified ontologies, despite being fundamental in AUI systems. These results suggest that, although there is a focus on contextual aspects of users and devices, the elements necessary to operationalize and standardize adaptations (Adaptive UI and Design Patterns) are being little explored.

As for *reuse* (RQ11), most works (around 61%) did not reuse any existing ontologies. The remaining 39.1% (nine papers) reused one or more existing ontologies. The FOAF ontology for user profiling was the most reused (#07, #09, and #13). The majority of studies opted for creating new ontologies, likely to address specific research needs or that researchers may be unaware of existing ontologies. While reuse can offer benefits such as reducing effort, enhancing interoperability, and building on established knowledge, challenges like a lack of awareness of existing ontologies or difficulties in adapting them to specific needs may explain this preference for creating new ontologies. These results may suggest that there is still a need to evolve mechanisms to stimulate and facilitate the reuse of ontologies.

Finally, regarding the *use of fundamental ontologies* (RQ12), it was observed that only one (4.3%) of the 23 papers addressed ontologies grounded on a fundamental ontology (specifically the Unified Foundational Ontology (UFO) Guizzardi (2005)). This result demonstrates that the use of fundamental ontologies to ground ontologies used in the development of AUI systems is not a common practice in the studies analyzed. This result suggests that there was little concern with good practices in ontology engineering, which may affect the quality of the ontologies produced, highlighting the importance of using fundamental ontologies. This result may also reveal an opportunity for greater rapprochement between the Software Engineering/Human-Computer Interaction communities, which are more concerned

with the development of AUI systems, and the Ontology community, which is concerned with the quality and application of ontologies.

Based on the panorama provided by the study results, in summary, the use of ontologies in the development of AUI systems has focused mainly on operational aspects, with an emphasis on implementing and automating UI adaptations based on context and user characteristics. Ontologies have demonstrated their value mainly in structuring and reasoning about the main elements of AUI systems. Furthermore, although the reuse of existing ontologies has been observed, the creation of new ontologies tailored to specific research needs remains prevalent, reflecting a lack of standardization and awareness of existing resources. The very rare use of foundational ontologies and little use of systematic methodologies highlight areas for exploration.

Overall, the findings suggest that ontologies have been useful in the development of AUI systems, and there is still room for improvement in how they have been developed and used. For example, we found that ontologies have been used primarily in operational aspects and in the implementation phase, which reveals an opportunity to expand their application to more conceptual phases of development, such as requirements specification, modeling, and design, where they could aid in understanding the domain and defining the structural model of the system. We also noted that, although reuse of existing ontologies has been observed, the creation of new ontologies is still prevalent, highlighting the need to further promote the reuse of existing ontologies, reduce the lack of standardization, and low awareness of available resources. The adoption of foundational ontologies has also been a limitation (only one of the analyzed ontologies is grounded on a foundational ontology), suggesting a need for improvement in the quality and conceptual rigor of the produced ontologies. Furthermore, the limited use of systematic approaches that guide the use of ontologies in the development of AUI systems indicates an opportunity to integrate ontologies into more structured methodologies. Finally, the lack of detailed documentation of ontologies in many studies hinders their understanding, application, and reuse, and suggests a need for ways to promote the specification and publication of ontologies related to AUI.

6. Study Limitations

Although the ambition of a systematic mapping study is to summarize all relevant research in an area, different sets of publications can be obtained due to several decisions and judgments taken Wohlin et al. (2012). Thus, as with any study, the study presented in this paper has some limitations that must be considered with the results. In addition, some challenges can reach the researchers during a systematic mapping study, such as how to select a comprehensive and relevant source of publications, how to consistently apply the inclusion/exclusion criteria, how to classify data, and how to interpret them Costa et al. (2021). In this study, we experienced these challenges and carried out some actions aiming at minimizing their influence on the results.

Publication selection and data extraction were performed by the first author. To reduce subjectivity, other researchers reviewed a sample of the extracted data. Another limitation is related to the classifications made for categorizing data. Some categories were established during data extraction based on data provided by the analyzed publications (e.g., categories used in RQ5). Classification schemes and data categorization were conducted by the first author and reviewed by other researchers. However, determining how publications fit into these categories involves a significant degree of judgment. Thus, different results could be obtained by other researchers.

Another limitation is the lack of detailed information about the ontologies in the papers. The papers primarily focused on the AUI system (or its UI) development, omitting details about the ontologies

employed. This omission may have limited the understanding of the approaches and techniques used and influenced the results. Additionally, several works followed their own notations and conventions to present the ontologies. This lack of consistency poses difficulties in understanding and comparing the findings of different papers.

Applying the search string to the digital libraries' engine can also interfere with the returned publications. First, because it is necessary to adapt the string syntax. The adaptation may result in a different search string. The string used in the study was adapted to each digital library, and we double-checked the adapted string and ran tests to ensure it was correct. Second, because of problems or changes in the engines. During the study, we did not notice changes in the engines of the used digital libraries.

Terminological problems in the search strings may have led to missing some primary studies. To minimize these problems, we performed previous simulations in the selected databases. Another limitation concerns the scope of the search. We decided not to search any specific conference proceedings, journals, or grey literature (e.g., technical reports and works in progress). Thus, we have considered only papers indexed by the selected digital libraries. The exclusion of these other sources makes the SMS more repeatable, but possibly some valuable papers may have been missed. Although the digital libraries adopted represent a comprehensive source of publications, the exclusion of other sources and the fact that we did not perform snowballing may have omitted some valuable publications from our analysis.

7. Final Considerations

This paper presented a systematic mapping study in which we analyzed 165 publications and identified 30 papers on the subject. After grouping complementary works, 23 works were analyzed. 12 research questions were defined to investigate the distribution of publications over the years and types of vehicles, the research types, and the following aspects: (i) the use of ontologies to aid in developing AUI systems and (ii) the used ontologies. Concerning (i), the RQs investigated the purposes for using ontologies, the software development phases they are applied in, how they have been used, the UI adaptations they support, the systematic development methods used, and the aspects considered to adapt the UI. Regarding (ii), the RQs investigated the types of ontologies used, the languages for their representation, the domains they cover, the reuse adoption, and the use of foundational ontologies.

Concerning *the use of ontologies to aid in developing AUI systems*, the results revealed that ontologies have primarily been used to provide a conceptual foundation, organizing elements such as user profiles, devices, and context of use, while supporting dynamic UI adaptations through semantic reasoning. Their application has focused mainly on the implementation phase. Furthermore, their potential for systematic and standardized use is underexplored. Regarding *the used ontologies*, the findings showed a predominance of operational ontologies, indicating that the utilization of ontologies has been predominantly centered on the operational aspects of AUI system development; OWL has been the most used language; Context of Use, Devices, and User Profile have been the domains covered more often by the ontologies, reuse has not been prominent but some works have done it, and using foundational ontologies to ground the ontologies has not been a concern.

During the analysis of publications in this study, we faced difficulties in extracting information about the ontologies, as it was either incomplete or implicit in the paper. This lack of information made it challenging to understand the existing ontologies and hindered data extraction. We believe that the community would benefit if the proposed ontologies were presented completely (even if through supplementary material). Additionally, it is fundamental to improve the visibility and access to the ontologies and their documentation on the web.

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